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LINEAR PROGRAMMING II

Linear Programming II
Duality, Shadow Prices & Sensitivity



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Today: Every LP Has a Twin

PART I · DUALITY

- The mystery numbers in Week 3's final tableau
- Shadow prices — the value of one more unit
- Building the dual LP (a pricing problem)
- Weak & strong duality; certificates
- Complementary slackness
- The primal–dual outcome matrix

PART II · PRICES IN PRACTICE

- Sensitivity: how far do prices hold?
- Objective ranging: robust plans
- Simplex vs interior-point methods
- Tooling: PuLP / OR-Tools / HiGHS
- Cases: data-center power · network bottlenecks
- Lab 4

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Week 3's Final Tableau Knew More Than x*

FINAL TABLEAU (Week 3, job-mix LP) — row 0 only

```
      x1  x2  s1  s2  s3 | rhs
 z     0   0   0 1.5 1.0 |  36
             ^^^ ^^^     the 'mystery numbers'
```

- **Last week: row 0 ≥ 0 was just the STOP sign. But the entries over the slacks — 0, 1.5, 1 — are data.**
- One number per RESOURCE: accelerator → 0, memory → 1.5, CPU → 1.
- Claim of the day: these are prices — exactly what one extra unit of each resource is worth.
- **Today we build the theory that says so — and shows it is no accident.**

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Shadow Prices: the Value of One More Unit

Definition: the shadow price y_i of constraint i = the increase in z* per unit increase in b_i (all else fixed).

Memory $y_2 = 1.5$

+1 block → z* rises 36 → 37.5.

Worth paying up to 1.5 k₺ to rent a block.

CPU $y_3 = 1$

+1 core → z* rises 36 → 37.

Positive, but memory is the better buy.

Accelerator $y_1 = 0$

+1 quota unit → z* unchanged.

2 units already idle — more is worthless.

⚠ *Marginal, not average — and only valid while the same constraints stay binding (ranges: Part II).*

The Dual LP: a Pricing Problem

Primal — the producer

- **max $z = 3x_1 + 5x_2$**
- $x_1 \le 4$ (accelerator)
- $2x_2 \le 12$ (memory)
- $3x_1 + 2x_2 \le 18$ (CPU)
- $x_1, x_2 \ge 0$
- Chooses a production plan x.

Dual — the resource pricer

- **min $w = 4y_1 + 12y_2 + 18y_3$**
- $y_1 + 3y_3 \ge 3$ (covers batch)
- $2y_2 + 2y_3 \ge 5$ (covers interactive)
- $y_1, y_2, y_3 \ge 0$
- One price y_i per RESOURCE; one constraint per PRODUCT.
- Cheapest prices that out-pay every product.

Recipe: constraint ⇄ variable · b ⇄ objective · c ⇄ rhs · max, ≤ ⇄ min, ≥ · A becomes A^T.

Why Prices Bound Profit: One Chain of Inequalities

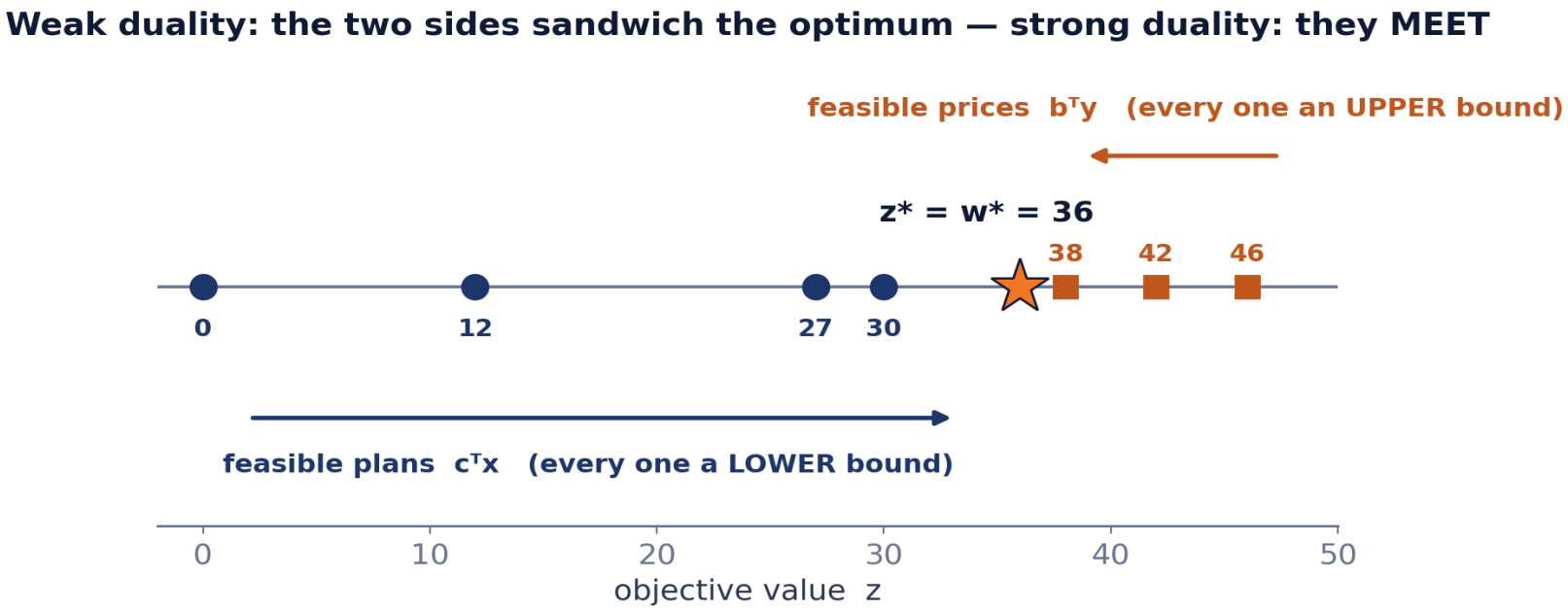
- **Take ANY feasible plan x and ANY non-negative prices y that cover every product (dual-feasible).**
- Price out the plan two ways — through the products, or through the resources:

$$\textbf{profit} = c^T x \leq (A^T y)^T x = y^T (Ax) \leq y^T b = \textbf{cost of the endowment}$$

① *every product's profit is covered by its priced recipe* ② *the plan uses no more than the endowment*

- Step ① uses dual feasibility ($A^T y \geq c$) and $x \geq 0$; step ② uses primal feasibility ($Ax \leq b$) and $y \geq 0$.
- **So every feasible price vector caps every feasible plan: no plan can out-earn any honest pricing.**

Weak Duality: a Sandwich of Bounds



Weak duality (theorem)

$c^T x \le b^T y$ for every feasible x and y

What it buys you

Any dual-feasible y certifies “no plan beats $b^T y$” — and if $c^T x = b^T y$, BOTH are provably optimal.

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Strong Duality: the Sandwich Closes

Theorem: **if an LP has an optimal solution, so does its dual, and $z^* = w^*$. The duality gap is ZERO.**

- **Job-mix check: $y^* = (0, 1.5, 1) \rightarrow w = 4 \cdot 0 + 12 \cdot 1.5 + 18 \cdot 1 = 36 = z^*$. ✓**
- The optimal prices exist, and they are exactly the shadow prices.
- Proof sketch: simplex terminates with row 0 ≥ 0 — those entries ARE a dual-feasible y with equal objective.

Where the dual hides

Solve the primal with simplex and the dual solution appears FREE OF CHARGE: y^* = final row 0 over the slack columns — Week 3's mystery numbers. Every LP solve is secretly two solves.

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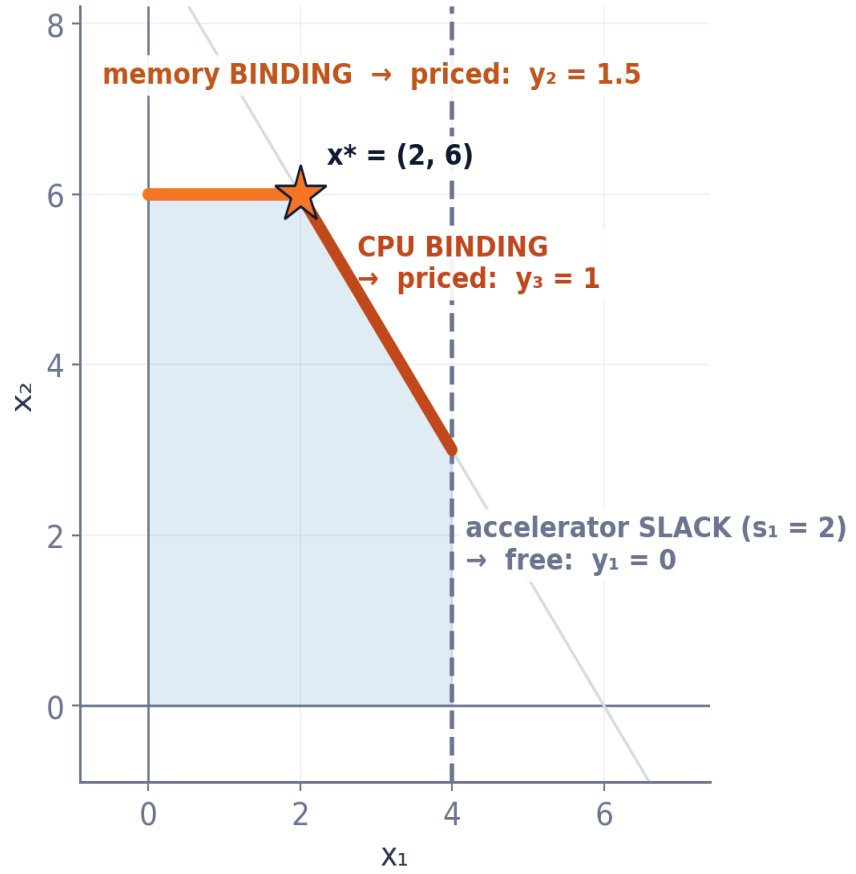
Complementary Slackness: Prices ⇄ Scarcity

- **At optimality, for every resource i: $y_i \cdot s_i = 0$.**
- Spare capacity ($s_i > 0$) ⇒ price zero — abundance is free.
- Positive price ($y_i > 0$) ⇒ resource fully used — value means scarcity.
- Mirror for products: $x_j > 0$ ⇒ its dual constraint is tight (fairly priced).

Why it's useful

It pins the solutions together: guess which constraints bind and complementary slackness hands you y* from x* (or back) by solving small linear equations — the lab does exactly this.

Complementary slackness: only binding constraints carry a price



The Primal–Dual Outcome Matrix

PRIMAL STATUS	DUAL STATUS	WHY
Primal optimal	**Dual optimal — and z* = w***	*the normal case: both solved at once*
Primal unbounded	**Dual INFEASIBLE**	*no finite prices can cap infinite profit*
Primal infeasible	**Dual unbounded (or infeasible)**	*with no plans to cap, prices can race to −∞ cost*
Both infeasible	**possible — but rare**	*contradictory constraints on both sides*

Debugging value: a solver saying "dual infeasible" is telling you your model is unbounded — go find the missing constraint.

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PART II

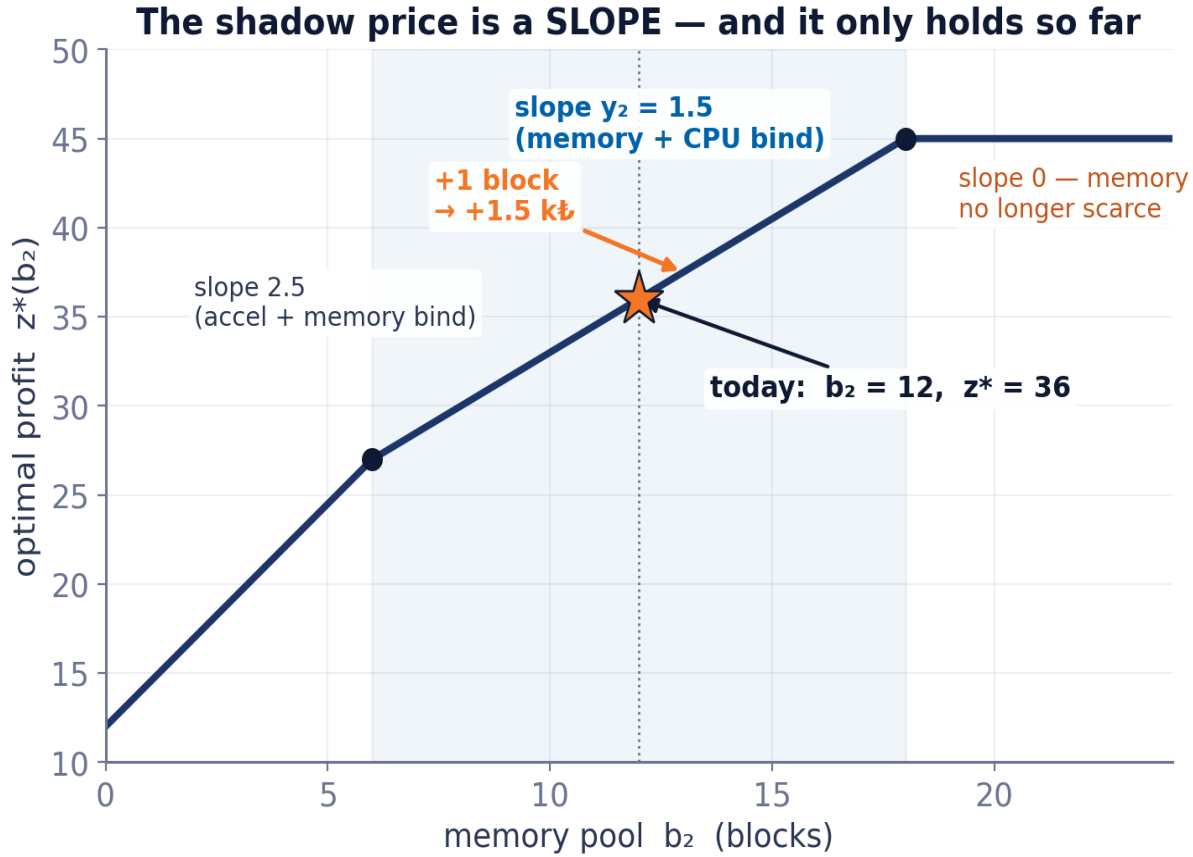
Prices in Practice

Sensitivity ranges, interior-point methods, real tooling & two engineering cases.

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How Far Does a Price Hold? RHS Ranging

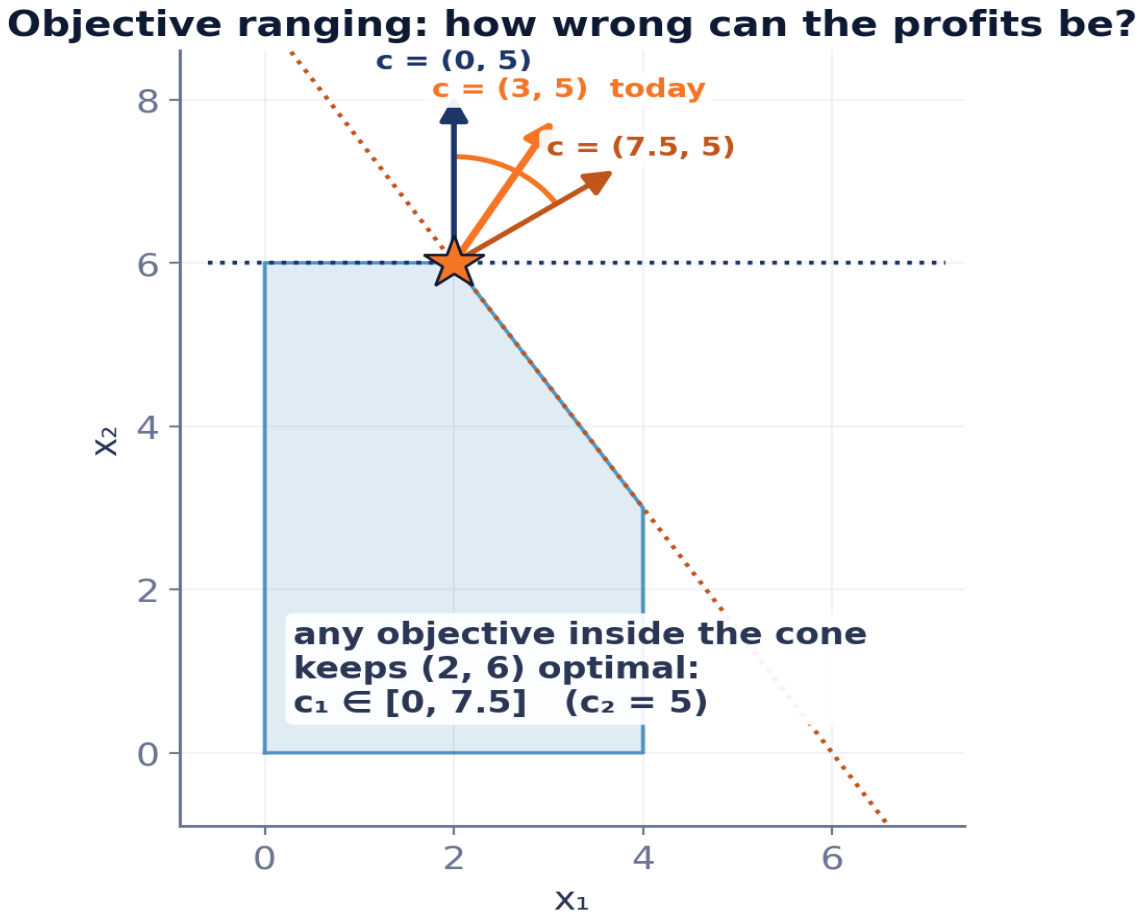
- **$z^*(b_2)$ is piecewise linear — the slope IS the shadow price.**
- Memory's price 1.5 holds for $b_2 \in [6, 18]$ — while {memory, CPU} stay binding.
- Below 6: a different basis binds — the price jumps to 2.5.
- Above 18: memory stops binding — its price collapses to 0.
- **Kinks = basis changes. Never extrapolate a shadow price past its range.**



Solvers report the range with the price — in the lab you'll SWEEP b_2 and draw this curve yourself.

How Wrong Can the Profits Be? Objective Ranging

- **Profit estimates are guesses. Does the PLAN survive if they're off?**
- Rotating c keeps (2, 6) optimal while c stays inside the normal cone of the vertex.
- Here: batch profit can be anything in [0, 7.5] (at $c_2 = 5$) — the plan is ROBUST.
- Leave the cone → a neighboring vertex takes over (alternative optima exactly at the boundary).
- Same z*-sensitivity logic, applied to c instead of b.



Manager's question answered: "our batch-profit estimate is ±30% — do we need to re-plan?" No.

Fine Print: When Price-Reading Goes Wrong

Degeneracy → ambiguous prices

At a degenerate optimum (Week 3!) the dual can be NON-UNIQUE: different solvers report different, equally valid shadow prices. Ranges shrink to zero.

Marginal ≠ total

A price values the NEXT unit only. 'Memory is worth 1.5/block' does not price the whole pool — integrate along the ranging curve instead.

One-at-a-time only

Ranging holds one coefficient fixed at a time. Changing b_2 and b_3 together can cross a kink neither would alone — re-solve for joint what-ifs.

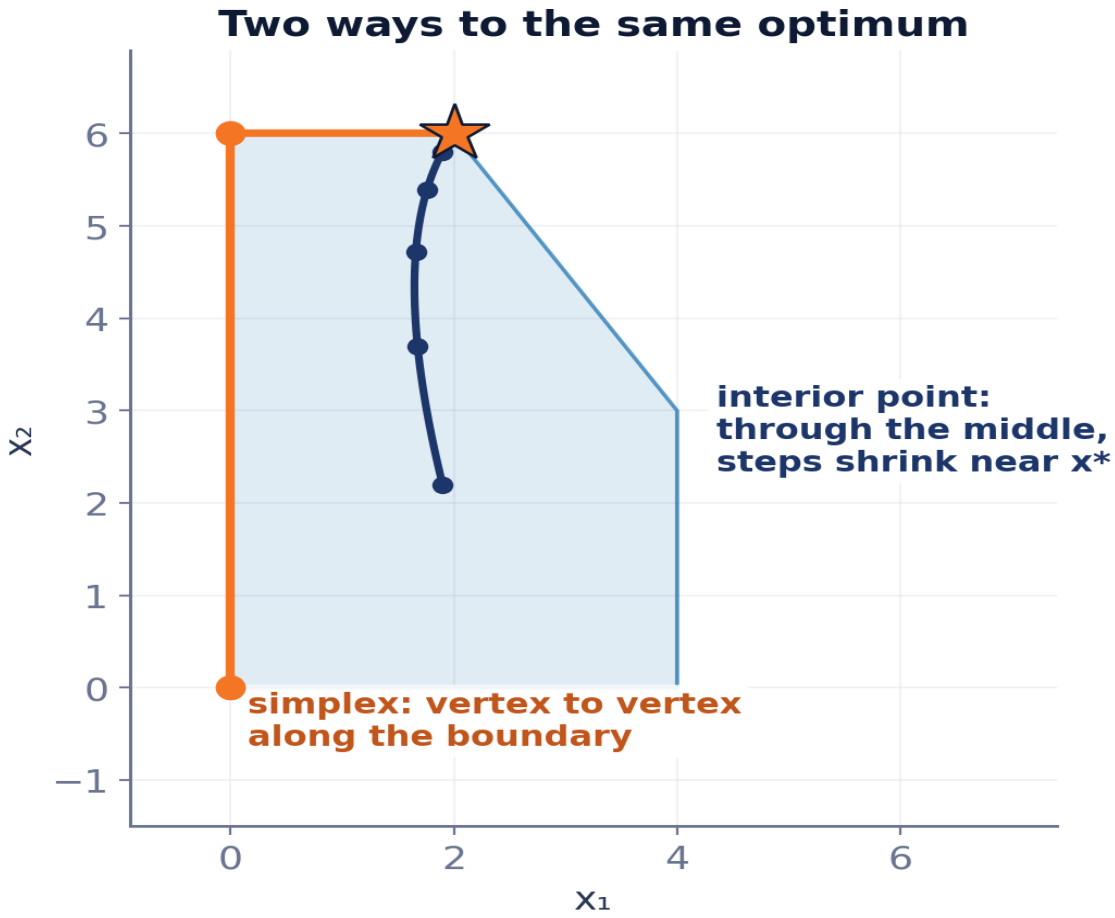
Integer models break it

Duals come from the LP relaxation. After branch-and-bound (Week 12) 'shadow prices' of an ILP are not well-defined — treat solver output with care.

Duals are the most-misread numbers in optimization practice — these four cover 90% of the accidents.

Simplex vs Interior-Point

- **Simplex: walks vertices on the boundary.**
- Exact vertex answers, warm-startable, exponential worst case.
- **Interior-point (barrier): smooth steps through the interior, following the 'central path'.**
- Polynomial worst case (Karmarkar 1984); shines on huge sparse LPs.
- Both produce the DUAL as they go — IPMs are literally primal-dual methods.
- **Production solvers race simplex AND barrier in parallel, take the first winner (then 'crossover' to a vertex).**



In Practice: Model in PuLP, Read Duals for Free

- **Modeling layers (PuLP, OR-Tools, Pyomo) write the math; solvers (CBC, HiGHS, Gurobi) do the work.**
- Name your constraints — every dual and slack is then one attribute away. (scipy: result.ineqlin.marginals)

```
import pulp as pl
m  = pl.LpProblem("job_mix", pl.LpMaximize)
x1 = pl.LpVariable("batch", lowBound=0);  x2 = pl.LpVariable("interactive", lowBound=0)
m += 3*x1 + 5*x2                                  # objective
m += x1 <= 4, "accel";  m += 2*x2 <= 12, "memory";  m += 3*x1 + 2*x2 <= 18, "cpu"
m.solve(pl.PULP_CBC_CMD(msg=0))
print(pl.value(m.objective))                      # 36.0
print({n: (c.pi, c.slack) for n, c in m.constraints.items()})
# {'accel': (0.0, 2.0), 'memory': (1.5, 0.0), 'cpu': (1.0, 0.0)}   <- y* and s*
```

Complementary slackness, live: every pair (pi, slack) has a zero — read the bottleneck ranking straight off.

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Case: Sizing a Data-Center Pod

The model

$$\max\ 30x_1 + 20x_2$$

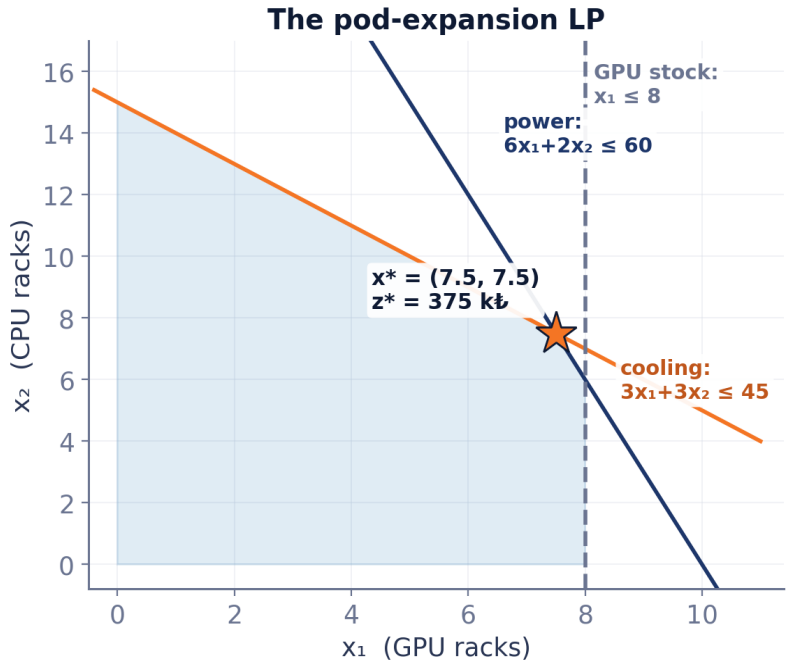
$6x_1 + 2x_2 \le 60$ (power)

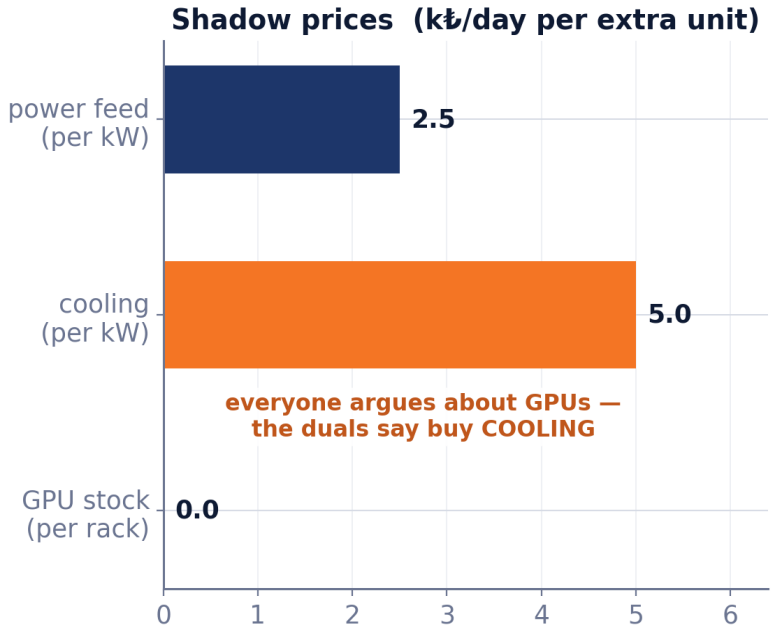
$3x_1 + 3x_2 \le 45$ (cooling)

$x_1 \le 8$ (GPU stock)

$x_1, x_2 \ge 0$

x_1 GPU racks · x_2 CPU racks · profit kŁ/day





The duals overturn the meeting-room debate: GPU stock is priced at ZERO (half a rack sits unused) — while every kW of cooling is worth 5 kŁ/day, twice the power feed. Buy chillers, not GPUs.

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The Duality Dictionary

CONCEPT	PRIMAL (production)	DUAL (pricing)
One resource	a row: $a_i^T x \le b_i$	a price: $y_i \ge 0$
One product	a column: $x_j \ge 0$	a covering row: $(A^T y)_j \ge c_j$
The data swap	profits c in the objective	capacities b in the objective
At the optimum	binding rows / slack rows	priced ($y_i > 0$) / free ($y_i = 0$)
Where it hides	x* in the basis & rhs	y* in final row 0 over the slacks

Weak duality: plans ≤ prices. Strong duality: they meet. Complementary slackness: pair by pair, one of each is zero.

WRAP-UP

Summary & Looking Ahead

KEY TAKEAWAYS

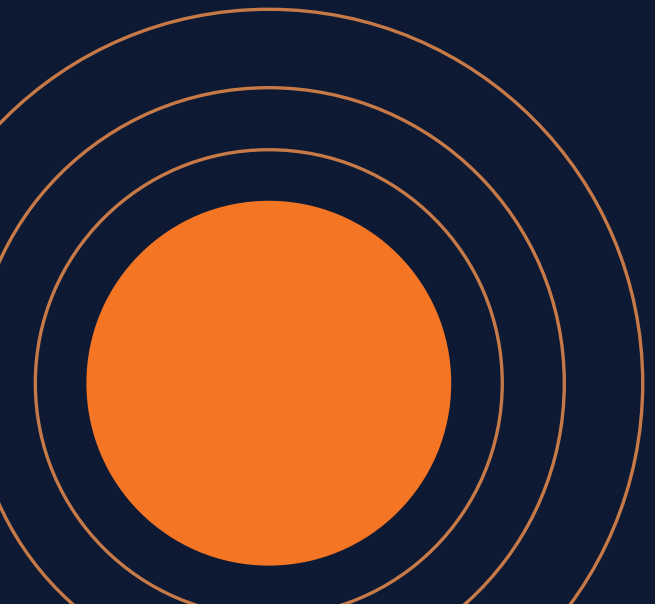
✓ Every LP has a dual: production ⇄ pricing, and solving one solves both.

✓ Weak duality bounds; strong duality closes the gap: $z^* = w^*$.

✓ Complementary slackness: only scarce resources carry a price.

✓ Shadow prices are slopes — valid only within their ranges.

✓ In practice: name constraints, read .pi/.slack, respect the fine print.



BEFORE NEXT WEEK

- Read Bertsimas & Tsitsiklis Ch. 4 · B&V §5.1–5.5 · problems: Winston Ch. 5–6
- Türkçe: Taha, Yöneylem Araştırması (dualite böl.) · Pınar, Doğrusal Optimizasyon notları

NEXT WEEK

Unconstrained Nonlinear Optimization I — gradient descent, line search & convergence: Week 2's calculus becomes an algorithm.